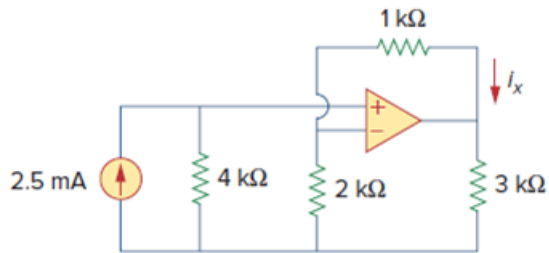


Problem

Refer to the op amp circuit in Fig. 5.71. Calculate i_x and the power absorbed by the 3-k Ω resistor.

Figure. 5.71:

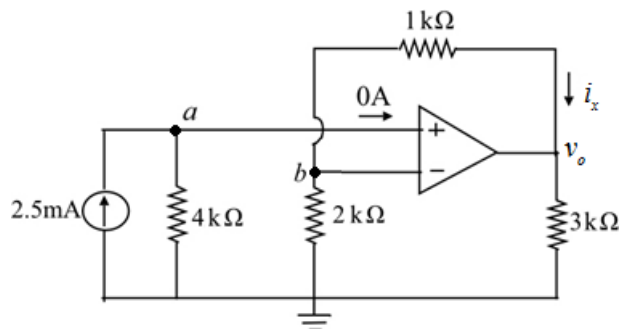


Step-by-step solution

Step 1 of 5

Refer to Figure 5.71 in the text book for the op-amp circuit.

Redraw the circuit by marking nodes.



Step 2 of 5

Apply Kirchhoff's current law at node a .

$$2.5 \text{ mA} = \frac{v_a}{4 \text{ k}\Omega} + 0$$

$$v_a = 10 \text{ V}$$

According to virtual ground concept, the voltage at the non-inverting input is equal to the voltage at the inverting input of the op-amp.

$$v_a = v_b = 10 \text{ V}$$

Step 3 of 5

Determine the current through $2\text{ k}\Omega$ resistor.

$$\begin{aligned}i_{2\text{ k}\Omega} &= \frac{v_b}{2\text{ k}\Omega} \\&= \frac{10}{2\text{ k}\Omega} \\&= 5\text{ mA}\end{aligned}$$

Apply Kirchhoff's current law at node b .

$$\begin{aligned}i_x &= -i_{2\text{ k}\Omega} \\&= -5\text{ mA}\end{aligned}$$

Therefore, the current i_x is $\boxed{-5\text{ mA}}$.

Step 4 of 5

The output of a noninverting amplifier is,

$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_a$$

Substitutes $1\text{ k}\Omega$ for R_f , $2\text{ k}\Omega$ for R_1 and 10 V for v_a .

$$\begin{aligned}v_o &= \left(1 + \frac{1\text{ k}\Omega}{2\text{ k}\Omega}\right)(10) \\&= 15\text{ V}\end{aligned}$$

Step 5 of 5

Determine the current through $3\text{ k}\Omega$ resistor.

$$\begin{aligned}i_{3\text{ k}\Omega} &= \frac{v_o}{3\text{ k}\Omega} \\&= \frac{15}{3\text{ k}\Omega} \\&= 5\text{ mA}\end{aligned}$$

Determine the power absorbed by $3\text{ k}\Omega$ resistor.

$$\begin{aligned}P_{3\text{ k}\Omega} &= v_o i_{3\text{ k}\Omega} \\&= (15\text{ V})(5\text{ mA}) \\&= 75\text{ mW}\end{aligned}$$

Thus, the power absorbed by $3\text{ k}\Omega$ resistor is $\boxed{75\text{ mW}}$.